

Project Initialization and Planning Phase

Date	15 March 2026
Team ID	TM-OPTICROP-04
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love. A well-articulated customer problem statement allows you and your team to find the ideal solution for your customers' challenges. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

Project Problem Statements (OptiCrop Engine):

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A local traditional farmer looking to transition to science-backed farming.	Select and cultivate the most suitable crop for the upcoming season to maximize harvest yield and revenue.	I lack precise insights into my soil's true NPK nutrient breakdown and local microclimate behaviors.	Traditional methods rely primarily on historical guesswork, which frequently fails due to volatile climate changes.	Anxious and uncertain about seed financial investments and seasonal income reliability.
PS-2	An eco-conscious farmer seeking resource efficiency and sustainable soil health.	Assess environmental suitability and optimize chemical fertilizer inputs for a specific desired crop.	I do not know the compatibility margins between my current soil metrics (like pH and rainfall) and the target	There are no localized tools to instantly evaluate soil compatibility metrics before purchasing inputs.	Frustrated by resource wastage and concerned about accelerating soil degradation over time.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
			crop's stress thresholds.		
PS-3	An agricultural researcher or regional policymaker.	Analyze macro crop-environment interactions to formulate regional data-driven sustainable strategies.	Aggregating and identifying hidden trend patterns from fragmented multi-dimensional environmental profiles is highly manual.	Existing software systems fail to combine NPK matrix records with scalable diagnostic visualization dashboards.	Restricted by slow evidence gathering, making it hard to deploy precise, proactive regional support policies.